AMAZON S3: OBJECT STORAGE AND MANAGEMENT

TYPES OF STORAGE

1. Simple Storage Service (S3)
2. Elastic File System (EFS)
3. Elastic Block Store (EBS)
4. FXS

DIFFERENCE BETWEEN OBJECT STORAGE AND BLOCK STORAGE

Block Storage

1. Block storage is suitable for transitional databases, random read/write loads and structured database storage.
2. Block storage divides the data to be stored in evenly sized blocks called data chunks for instance; a file can be split into evenly sized blocks before it is stored.
3. Data blocks stored in block storage would not contain metadata. (Data created, data modified, content type etc.)
4. Block storage only keeps the address (index number) where the data blocks are stored, it does not care what is in that block, just how to retrieve it when required.

Object Storage

1. Object storage stores the files as a whole and does not divide them.
2. In object storage an object is: the file/ data itself, its Meta data, object global unique ID.
3. The object global unique ID is a unique identifier for the object (can be the object name itself) and it must be unique such that it can be retrieved disregarding where it's physical storage location is.
4. Object storage cannot be mounted as a drive.
5. Example of object storage solutions are Dropbox, AWS S3, Facebook.

SIMPLE STORAGE SERVICE:-(S3)

1. Amazon s3 is a cloud-based object-oriented storage for storing any type of data any form of data and retrieve the data anytime, anywhere on the internet
2. S3 is object-based storage So You cannot install operating system on S3.

3. S3 has a distributed data store architecture where objects are redundantly stored in multiple locations. (minimum 3 locations in same region)
4. Data is stored in bucket.
5. A bucket is a flat container of objects. (you cannot create nested buckets)
6. Maximum capacity of a bucket is 5TB.
7. Bucket ownership is non transferrable.
8. S3 bucket is region specific.
9. You can have up to 100 buckets per account. (may expand on request)

S3 BUCKET NAMING RULES

1. S3 bucket names must be globally unique across all AWS accounts and regions.
2. Bucket names cannot be changed after creation.
3. If a bucket is deleted, its name becomes available again and can be used by any AWS account.
4. Bucket names must be between 3 and 63 characters long.
5. Bucket names are part of the URL used to access the bucket, so they must follow DNS naming conventions.
6. Bucket names must be a combination of labels (words) separated properly, following DNS format.
7. Bucket names can contain lowercase letters, numbers, and hyphens (-) only.
8. Uppercase letters and special characters are not allowed.
9. Bucket names must not be formatted as an IP address (e.g., 192.168.1.1).
10. Bucket names must start and end with a lowercase letter or number.
11. By default, all buckets and objects are private, and only the bucket owner has access unless permissions are changed.

CREATE S3 BUCKET & ADD OBJECTS IN THE BUCKET

1. search s3 bucket
2. create a bucket
    1. name ----------> amazon-s3-bucket
    2. bucket type ----> general purpose bucket
    3. public access ---> unblock all public access
    4. create a bucket
3. click on the bucket
    1. click on upload -----> add files -----> select file.txt
    2. upload

HOW TO DELETE A S3 BUCKET

1. First delete all the files from these bucket
2. Then delete empty bucket

S3 PROPERTIES

1. BUCKET VERSIONING

1. Bucket versioning is a S3 bucket sub resource used to protect against from accidental object/data deletion or overwrites.
2. Versioning can also be used for data retention and archive purposes.
3. Once versioning is enabled on a bucket, it cannot be disabled, but it can be suspended.
4. When versioning enabled, it protects both existing and new objects by maintaining their versions whenever they are updated.
5. Updating objects refers to PUT, POST, COPY, DELETE actions on objects.
6. When versioning is enabled and you try to delete an object a delete marker is placed on the object.
7. You can still view the object and delete the marker.
8. If you reconsider deleting objects you can delete the delete marker and the object will be accessible again.

PRACTICAL

1. create a bucket & enable the versioning
2. create a file & add data into the file
3. add this file into the s3 bucket
4. then one more time add the data into the file
5. and upload this file into the s3 bucket
6. then simply click on show versions you get the version for each data

---

2.ENCRYPTION IN S3

1. SSE S3 (Server-Side Encryption with S3)
   Default encryption managed by AWS

2. SSE KMS (Server-Side Encryption with KMS)
   Encryption managed using AWS Key Management Service for better control and auditing

3. DSSE KMS (Dual-Layer SSE-KMS)
   Provides dual-layer encryption using KMS for higher security

---

3. SERVER ACCESS LOGGING

1. Server access logging tracks who accessed your S3 bucket and what actions they performed.
2. It is useful for security analysis, troubleshooting, and tracking access patterns.
3. It captures details like requester, request time, IP address, operation, and response status.

---

4. TRANSFER ACCELERATION

1. Transfer Acceleration in Amazon S3 is used to upload and download data faster over long distances.
2. It uses Amazon CloudFront edge locations to transfer data through optimized network paths.
3. Data is first sent to the nearest edge location and then transferred to the S3 bucket.
4. It is useful when users are located far from the bucket region and need faster data transfer.

5.OBJECT LOCK

It is used to prevent objects from being deleted or modified for a fixed period of time (WORM – Write Once Read Many).

PRACTICAL

1. click on edit
2. click on object lock ---> enable
3. default retention mode -------> compliance
4. default retention ------> Enable ------> 90
5. save changes

6.EVENT NOTIFICATIONS

Event Notifications in Amazon S3 automatically trigger actions (like Lambda, SNS, SQS) when events such as object upload or delete occur.

7.REQUESTER PAYS

Requester Pays in Amazon S3 means the person who accesses the data (requester) pays for the data transfer and request costs instead of the bucket owner.

BUCKET ACL

1.go to permission section
2.click on object ownership
   1. object ownership -----> ACLs enabled
   2. bucket owner preferred
   3. save changes

3.click on the object ---> enter into the object
   1. click on permission
   2. Edit ACL
   3. Everyone -----> read access
   4. save changes

4.another way to directly get the ACL access to objects
   1. click on the objects
   2. click on action -------> make public using acl

BUCKET POLICY

1. A Bucket Policy in Amazon S3 is a JSON-based access control policy
2. It is used to grant or deny permissions to an entire bucket and all its objects

CREATE BUCKET POLICY & ADD

1. click on bucket policy
2. click on policy generator
   1. type of policy -------> s3 bucket policy
   2. effect ------> allow
   3. principle --> *
   4. actions ----> all
   5. ARN --------> add bucket ARN

6. Add statement
7. Generate policy -----> copy policy and paste in the bucket
8. save changes

Note:
When u add the policy in the bucket only add /* after bucket arn

---

8.STATIC WEB HOSTING

1. static web-hosting is a serverless hosting generally developers use this service to host their website serverless means without server-side processing.
2. it is affordable and scalable solution for hosting static website and its low traffic website provides high availability and durability to website.
3. Allows bucket to host a static website (HTML, CSS, JS)
4. Used for simple web apps

PRACTICAL

1. Download the css template from browser
2. add these template files in the s3 bucket
3. assign the public access to these files using ACL
4. go to s3 properties
5. static website hosting ------> edit
   1.static website hosting ----> enable
   2.index document -----> index.html
   3.save changes
6. copy the static website hosting URL & paste it on browser
